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MOUNTING BRACKET AND SERVER HAVING THE MOUNTING BRACKET

Abstract

A mounting bracket and a server having the mounting bracket are disclosed. The mounting bracket includes a supporting member, a locking member, and an unlocking member. The unlocking member can move between a locked position and an unlocked position, thereby causing deformation of the locking member, which enables the locking member to connect with or disconnect from an external component. The mounting bracket can connect the workpiece and the external component without the need for tools, which helps to reduce assembly costs and maintenance costs.

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Background/Summary

FIELD

[0001] The subject matter herein generally relates to servers, and more particularly, to a mounting bracket and a server having the mounting bracket.

BACKGROUND

[0002] When connecting a workpiece to external components, fasteners, such as bolts and screws, are needed, and tools for operating the fasteners are also needed, resulting in low assembly efficiency and inconvenient maintenance.

[0003] Therefore, there is room for improvement in the art.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Implementations of the present disclosure will now be described, by way of embodiments only, with reference to the attached figures.

[0005] FIG. 1 is a diagram showing an embodiment of a mounting bracket according to the present disclosure.

[0006] FIG. 2 is a diagram showing a partial structure of the mounting bracket of FIG. 1 in a certain state.

[0007] FIG. 3 is an enlarged view of a portion A in FIG. 2. FIG. 4 is an enlarged view of a circled portion B in FIG. 1.

[0008] FIG. 5 is a diagram showing a partial structure of the mounting bracket of FIG. 1 in another state.

[0009] FIG. 6 is a diagram showing an embodiment of a partial structure of a server according to the present disclosure.

[0010] FIG. 7 is an exploded view of the server in FIG. 6.

DETAILED DESCRIPTION

[0011] Some embodiments of the present application will be described in detail below with reference to the drawings. It is evident that the described embodiments are merely some, rather than all, of the embodiments of the present disclosure.

[0012] It should be noted that when a component is referred to as being “fixed on” or “mounted on” another component, it may be directly on the other component or there may also be an intervening component. When a component is considered to be “set on” another component, it may be in direct contact with the other component or there may also be an intervening component. As used herein, the term “and/or” includes all and any combinations of one or more of related listed items.

[0013] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by persons skill in the art. The terms used herein are only for the purpose of describing specific embodiments, and not intended to limit the embodiments of the present application.

[0014] In this application, descriptions such as “first”, “second” etc. are only used to distinguish between different objects, and should not be understood as indicating or implying relative importance, or as implicitly specifying the quantity, particular order, or a primary-secondary relationship of the technical features indicated.

[0015] The mention of “embodiments” in the present disclosure implies that the specific features, structures, or characteristics described in conjunction with the embodiments may be included in at least one embodiment of the present application. The appearance of this phrase in various locations

in the specification does not necessarily refer to the same embodiment and is not intended to be an independent or alternative embodiment that is mutually exclusive of other embodiments. The various following embodiments may be combined with each other in the absence of conflict. [0016] It should be noted that the thicknesses, lengths, widths, and other dimensions of various components shown in the figures of the embodiments of the present application, as well as the overall thickness, lengths, widths, and other dimensions of the integrated device, are for illustrative purposes only and should not constitute any limitation on the present application.

[0017] Some embodiments of the present application will be described in detail below with reference to the drawings.

[0018] Referring to FIGS. **1** and **2**, an embodiment of the present application provides a mounting bracket **100** including a supporting member **10**, a locking member **20**, and an unlocking member **30**.

[0019] The supporting member **10** is used to support a workpiece **200** (referring to FIG. **6**).

[0020] The locking member **20** is connected to the supporting member **10**, and can be connected to an external component **300** (referring to FIG. **6**) through a snap-fit manner.

[0021] The snap-fit between the locking member **20** and the external component **300** is simple, which is conducive to improving the efficiency of assembly and disassembly between the locking member **20** and the external component **300**. In at least one embodiment, the locking member **20** includes a latching block **21**, and the external component **300** defines a latching slot or includes a latching surface (not shown). The latching block **21** is used for engaging with the latching slot or latching surface. The structure of the latching block **21** is simple and easy to produce, which helps reduce manufacturing costs.

[0022] The unlocking member **30** is slidably connected to the supporting member **10**, and the unlocking member **30** may be stacked on a side of the locking member **20**. The unlocking member **30** can move between a locked position and an unlocked position to drive the locking member **20** to move or deform, thereby allowing the locking member **20** to switch between a locked state and an unlocked state. When the unlocking member **30** is in the locked position, the locking member **20** is connected to and locks the external component **300**. When the unlocking member **30** is in the unlocked position, the locking member **20** is disengaged from and unlocks the external component **300**. The locking member **20** can move or be deformed to disengage from the external component **300**, which simplifies the operation and eliminates the need for using tools, thereby improving the efficiency of disassembly of the locking member **20** and the external component **300** and reducing the maintenance cost of the workpiece **200**.

[0023] As shown in FIGS. **1**, **2**, and **5**, the unlocking member **30** can move along a first direction X to reach the unlocked position, and during the sliding process, the unlocking member **30** causes the locking member **20** to deform such that the locking member **20** can be disengaged from the external component **300**. The unlocking member **30** can also move along a second direction Y to reach the locked position, and during the sliding process, the locking member **20** rebounds to its original position or state, such that the locking member **20** can lock the external component **300**. The locking member **20** can deform or rebound means that the locking member **20** is made of an elastic material.

[0024] In some embodiments, the first direction X is opposite to the second direction Y. It should be noted that the movement direction of the unlocking member **30** is not limited to the first direction X and the second direction Y. In other embodiments, the unlocking member may have different movement directions. For example, the movement direction of the unlocking member **30** may be a direction perpendicular to the first direction X and the second direction Y. The movement direction of the unlocking member **30** may be varied according to the connection position of the external component **300** and the mounting bracket **100**. The following description is illustrated when the unlocking member **30** moves along the first direction X or the second direction Y.

[0025] In some embodiments, one of the locking member **20** and the unlocking member **30** defines

a mounting hole **22**, and the other of the locking member **20** and the unlocking member **30** has a protrusion **31**. As shown in FIGS. 2 and 5, in this embodiment, the locking member defines the mounting hole **22**, and the unlocking member **30** has the protrusion **31**. The protrusion **31** can be received in the mounting hole **22**. When the unlocking member **30** moves along the first direction X, the protrusion **31** moves towards the exterior of the mounting hole **22** along the first direction X (the protrusion **31** may partially or completely move out of the mounting hole **22**) to press against the portion of the locking member **20** near the mounting hole **22**, driving the locking member **20** to deform, thereby the locking member **20** can be disengaged from the external component **300**. When the unlocking member **30** moves along the second direction Y, the protrusion **31** moves towards the interior of the mounting hole **22** until the protrusion **31** is fully accommodated within the mounting hole **22**, and the locking member **20** rebounds to allow the latching block **21** to lock the external component **300**. The structure of the mounting hole **22** and the protrusion **31** is simple and easy to produce, which helps save manufacturing costs. Additionally, the simple connection between the mounting hole **22** and the protrusion **31** facilitates easier assembly of the locking member **20** and the unlocking member **30**, improving production efficiency of the mounting bracket **100**.

[0026] In some embodiments, the locking member **20** is integrally formed.

[0027] In some embodiments, as shown in FIGS. 2 and 5, the locking member **20** includes a first portion **210**, a second portion **220**, and a torsion spring (not shown in the figures). The first portion **210** is hinged to the second portion **220** and connected to the supporting member **10**. The second portion **220** is connected to the external component **300** through a snap-fit manner, and the torsion spring is connected to the first portion and the second portion. When the locking member **20** switches from the locked state to the unlocked state, the first portion **210** and the second portion **220** rotate relative to each other, and the torsion spring generates an elastic force. When the locking member **20** switches from the unlocked state to the locked state, the elastic force of the torsion spring enables the locking element **20** to return to the locked state. Since the locking member **20** includes the first portion **210**, the second portion **220**, and the torsion spring, when one of the above three components breaks, only the broken component is needed to be replaced, which is conducive to reducing the maintenance cost of the locking member **20**.

[0028] In other embodiments, the locking member **20** includes a first portion **210** and a second portion **220**. The first portion **210** is connected to the supporting member **10**, and the second portion **220** is hinged to the unlocking member **30**. The second portion **220** is connected to the external component **300** through a snap-fit manner. When the unlocking member **30** slides relative to the supporting member **10**, the unlocking member **30** pulls the second portion to rotate relative to the first portion, allowing the locking member **20** to switch between the unlocked state and the locked state. Since the locking member **20** includes the first portion **210** and the second portion **220**, when one of the above two components breaks, only the broken component is needed to be replaced, which is conducive to reducing the maintenance cost of the locking member **20**.

[0029] In some embodiments, when the workpiece **200** is placed on the supporting member **10**, the workpiece **200** may be connected to the supporting member **10** such as by adhesive bonding, snap-fit, insertion, magnetic adsorption, thereby enhancing the stability of the connection between the workpiece **200** and the external component **300**.

[0030] As shown in FIG. 1, in some embodiments, the supporting member **10** defines an installation groove **111**, and the workpiece **200** is at least partially accommodated in the installation groove **111**. The sidewalls of the installation groove **111** can limit the movement of the workpiece **200**, thereby improving the stability of the workpiece **200** on the supporting member **10**, and thus improving the stability of the connection between the workpiece **200** and the external component **300**.

[0031] In some embodiments, when the workpiece **200** is placed in the installation groove **111**, the workpiece **200** presses against the sidewalls of the installation groove **111**, thereby generating a damping force between the workpiece **200** and the sidewalls. Therefore, the possibility of the

workpiece **200** separating from of the installation groove **111** is reduced, thereby improving the stability of the workpiece **200** on the supporting member **10**, and thus improving the stability of the connection between the workpiece **200** and the external component **300**.

[0032] In some embodiments, an elastic material layer, such as a silicone layer or a rubber layer, is attached to the sidewalls of the installation groove **111**. When the workpiece **200** is placed in the installation groove **111**, the workpiece **200** presses against the elastic material layer, thereby generating the damping force between the workpiece **200** and the elastic material layer. The elastic material layer not only increases the damping applied to the workpiece **200** and improve the stability of the workpiece **200**, but also reduces the damages to the sidewalls of the workpiece **200**.

[0033] As shown in FIGS. **1**, **2**, and **5**, in some embodiments, the mounting bracket **100** further includes a fixing member **40**. The fixing member **40** is connected to the supporting member **10** and includes a limiting portion **41**. The limiting portion **41** is connected to a top surface of the workpiece **200** to press the workpiece **200** against the supporting member **10**, thereby improving the stability of the workpiece **200** on the supporting member **10**, and thus improving the stability of the connection between the workpiece **200** and the external component **300**.

[0034] Furthermore, the fixing member **40** further includes a fixing member body, and the limiting portion **41** is disposed at an end of the fixing member body. The height of the fixing member body is adjustable, so that the fixing member **40** can adapt to workpieces **200** of different heights, which is beneficial for making the mounting bracket **100** compatible with different usage scenarios.

[0035] Furthermore, the fixing member body includes an outer sleeve and an inner sleeve (not shown in the figures). The inner sleeve is connected to the limiting portion **41**, while the outer sleeve defines an accommodation space where the inner sleeve is slidably inserted. The outer sleeve also defines a limiting hole communicating with the accommodation space, and the inner sleeve includes a protruding button for engaging with the limiting hole. By altering the relative position between the inner sleeve and the outer sleeve, the height of the fixing member body can be adjusted. When the protruding button is accommodated within the limiting hole, the relative position between the outer sleeve and the inner sleeve is fixed. This configuration results in a simple and reliable structure for the fixing member body.

[0036] As shown in FIG. **6**, in some embodiments, the workpiece **200** may be in contact with the supporting member **10**. The relative position between the workpiece **200** and the supporting member **10** is fixed by the external component **300**. For instance, the external component **300** may include a first connecting portion and a second connecting portion, and the locking member **20** is connected to the first connecting portion. After positioning the workpiece **200** on the supporting member **10**, the second connecting portion is connected to the first connecting portion. The second connecting portion is in contact with the workpiece **200** and presses the workpiece **200** against the supporting member **10**. Therefore, there is no need to fix the workpiece **200** to the supporting member **10**, simplifying the operational steps and enhancing the efficiency of assembling and disassembling the workpiece **200** and the supporting member **10**.

[0037] In some embodiments, when the workpiece **200** is positioned on the supporting member **10**, the workpiece **200** may be electrically connected to the locking member **20**. When the locking member **20** engages with the external component **300**, the locking element **20** is also electrically connected to the external component **300**. This facilitates the electrical connection between the workpiece **200** and the external component **300**, reducing the complexity of wiring.

[0038] In some embodiments, the workpiece **200** includes a first PIN, and the locking member **20** includes a first contact point. When the workpiece **200** is positioned on the supporting member **10**, the first PIN is in contact with and electrically connected to the first contact point, thereby electrically connecting the workpiece **200** to the locking member **20**. The locking member **20** also includes a second contact point electrically connected to the first contact point, and the external element **300** includes a second PIN. When the locking member **20** engages with the external component **300**, the second PIN is in contact with and electrically connects to the second contact

point. The first PIN, the first contact point, the second PIN, and second contact point are structurally simple and easy to manufacture and assemble.

[0039] In some embodiments, the first contact point and the second contact point are formed by elastic metal sheets disposed on the locking member **20**. The elastic metal sheets can reduce the likelihood of the first PIN (the second PIN) disconnecting from the first contact point (the second contact point), thereby enhancing the stability of the electrical connection between the workpiece **200** and the external component **300**.

[0040] In some embodiments, a flexible wire is connected to the elastic metal sheet forming the first contact point and the elastic metal sheet forming the second contact point. This allows for easy repositioning of the flexible wire, reducing the complexity of wiring.

[0041] In some embodiments, when the workpiece **200** is positioned on the supporting member **10**, the workpiece **200** may be electrically connected to the carrier **10**. When the locking member **20** engages with the external component **300**, the supporting member **10** is electrically connected to the external component **300**. This facilitates the electrical connection between the workpiece **200** and the external component **300**, reducing the complexity of wiring.

[0042] In some embodiments, the workpiece **200** includes a third PIN, and the supporting member **10** includes a third contact point. When the workpiece **200** is positioned on the supporting member **10**, the third PIN is in contact with and electrically connected to the third contact point, thereby electrically connecting the workpiece **200** to the supporting member **10**. The supporting member **10** also includes a fourth contact point electrically connected to the third contact point, and the external component **300** includes a fourth PIN. When the locking member **20** engages with the external component **300**, the fourth PIN is in contact with and electrically connected to the fourth contact point. The third PIN, the third contact point, the fourth PIN, and the fourth contact point are structurally simple and easy to manufacture and assemble.

[0043] In some embodiments, the third contact point and the fourth contact point are formed by elastic metal sheets disposed on the supporting member **10**. The elastic metal sheets can reduce the likelihood of the third PIN (the fourth PIN) disconnecting from the third contact point (the fourth contact point), thereby enhancing the stability of the electrical connection between the workpiece **200** and the external component **300**.

[0044] In some embodiments, a flexible wire is connected to the elastic metal sheet forming the third contact point and the elastic metal sheet forming the fourth contact point. The position of the flexible wire can be easily changed, which reduces the difficulty of wiring.

[0045] It is easy to imagine that in the aforementioned embodiments, the positions of the first PIN and the first contact point, the second PIN and the second contact point, the third PIN and the third contact point, as well as the fourth PIN and the fourth contact point can be interchangeable to achieve the same effect.

[0046] As shown in FIGS. **1** and **6**, in some embodiments, the supporting member **10** includes a base **11** and a mounting portion **12**. The base **11** and the mounting portion **12** are separately configured. The base **11** is used to support the workpiece **200**, while the mounting portion **12** is used to install the locking member **20** and the unlocking member **30**. This allows for connecting the locking member **20** and the unlocking member **30** to the mounting portion **12** before attaching the mounting portion **12** to the base **11**, reducing assembly difficulty and saving assembly costs.

[0047] In other embodiments, the base **11** and the mounting portion **12** are integrally formed, which facilitates reducing the difficulty of manufacturing the supporting member **10** and saving costs.

[0048] In some embodiments, the supporting member **10** and the locking member **20** are integrally formed. The step of assembling the supporting member **10** and the locking member **20** are eliminated, thereby enhancing the assembly efficiency of the mounting frame **100**.

[0049] As shown in FIGS. **1** to **3**, in some embodiments, the mounting bracket **100** further includes a handle **50**. The handle **50** is rotatably connected to the unlocking member **30** and includes an arcuate surface **511** that is in contact with the mounting portion **12**. As shown in FIGS. **3** and **4**, the

arcuate surface **511** has a first contact area **5111** and a second contact area **5112**. From the first contact area **5111** to the second contact area **5112**, the radius of curvature of the arcuate surface **511** gradually changes. In this embodiment, the radius of curvature of the first contact area is greater than that of the second contact area. The handle **50** has a first positional state where the first contact area **5111** is in contact with the mounting portion **12** and a second positional state where the second contact area **5112** is in contact with the mounting portion **12**. The handle **50** can rotate between the first positional state and the second positional state. When the handle is in the first positional state, the unlocking member **30** is in an unlocked position. When the handle is in the second positional state, the unlocking member **30** is in a locked position. The handle **50** allows the operator to easily apply force to the unlocking member **30**, reduces the difficulty of moving the unlocking member **30**, enhances the efficiency of disassembly of the locking member **20** and the external component **300**, and thereby lowering the maintenance cost of the workpiece **200**.

[0050] As shown in FIG. 3, in some embodiments, the handle **50** includes a connecting portion **51**. The arcuate surface **511** is formed on the connecting portion **51**. One of the connecting portion **51** and the supporting member **10** defines a guide groove **512**, while the other of the connecting portion **51** and the supporting member **10** includes a guide protrusion **121**. In this embodiment, the connecting portion **51** defines the guide groove **512**, and the supporting member **10** includes the guide protrusion **121**. The guide protrusion **121** is received in the guide groove **512** to restrict axial movement of the handle **50**. The engagement between the guide protrusion **121** and the guide groove **512** enhances the stability of the rotation of the handle and optimize the user's operational experience.

[0051] As shown in FIGS. 1, 2, and 5, in some embodiments, the mounting bracket **100** further includes an elastic member **60** connecting the handle **50** and the supporting member **10**. When the handle **50** is in the first positional state, the elastic member **60** has a first deformation amount; when the handle **50** is in the second positional state, the elastic member **60** has a second deformation amount. The first deformation amount is greater than the second deformation amount. That is, the elongated length of the elastic member **60** in the first positional state is greater than the elongated length of the elastic member **60** in the second position state. During the transition of the handle **50** from the first positional state to the second positional state, the elastic member **60** applies a tensile force to the handle **50**, enabling the handle **50** to move towards the second positional state under the tensile force without requiring the user to apply force continuously, thereby optimizing the user's experience. Additionally, under the action of the elastic member **60**, the handle **50** and the supporting member **10** can jointly clamp the workpiece **200**, which helps improve the stability of the workpiece **200** on the supporting member **10**. Furthermore, when the handle **50** moves towards the first positional state, the handle **50** drives the unlocking member **30** to move towards the unlocked position, so that while the handle **50** releases the restriction on the workpiece **200**, the locking member **20** disconnects from the external component **300**. At this point, the workpiece **200** can be removed from the supporting member **10**, and the locking member **20** can be removed from the external component **300**, which improves the efficiency of disassembly of the workpiece **200**, the mounting bracket **100**, and the external component **300**.

[0052] In some embodiments, the elastic member **60** is connected to the unlocking member **30** and the supporting member **10**, thereby reducing the length of the elastic member **60** and saving costs.

[0053] In some embodiments, the locking member **20** has a first inclined surface (not shown in the figures), and the unlocking member **30** has a second inclined surface (not shown in the figures) that can mate with the first inclined surface. When the unlocking member **30** moves from the locked position to the unlocked position, the second inclined surface moves along the first inclined surface and drives deformation of the locking member **20**.

[0054] In some embodiments, one of the supporting member **10** and the unlocking member **30** includes a guide post **122**, and the other defines a guide hole **32**. As shown in FIGS. 2 and 5, in this embodiment, the supporting member **10** includes the guide post **122**, and the unlocking member **30**

defines the guide hole **32**. The guide post **122** passes through the guide hole **32**, and the unlocking member **30** is slidably connected to the supporting member **10** through the engagement of the guide post **122** and the guide hole **32**. The simple structure of the guide post **122** and the guide hole **32** reduces the difficulty of assembling the unlocking member **30** and the supporting member **10**, and improve production efficiency of the mounting bracket **100**. At the same time, the engagement of the guide post **122** and the guide hole **32** can guide the sliding direction of the unlocking member **30**, thus improving the stability of the unlocking member **30** during movement.

[0055] As shown in FIGS. **1** and **7**, in some embodiments, the supporting member **10** defines a positioning hole **112**, and the external component **300** includes a positioning post **3011**. The positioning post **3011** can be engaged in the positioning hole **112** to locate the supporting member **10** and quickly find the position of the mounting bracket **100** relative to the external component **300**, thus improving the efficiency of connecting the mounting bracket **100** with the external component **300**.

[0056] In some embodiments, the supporting member **10** includes the positioning post **3011**, and the external component **300** defines the positioning hole **112**. The positioning post **3011** can be engaged in the positioning hole **112** to locate the supporting member **10** and quickly find the position of the mounting bracket **100** relative to the external component **300**, thus improving the efficiency of connecting the mounting bracket **100** with the external component **300**.

[0057] As shown in FIG. **7**, in some embodiments, the external component **300** defines a latching hole **3012**. After the latching block **21** passes through the latching hole **3012**, the latching block **21** contacts with the external component **300**. At this point, the surface of the external component **300** in contact with the latching block **21** forms the aforementioned latching surface.

[0058] In other embodiments, the locking member **20** defines the latching slot, and the external component **300** includes the latching block **21** for engaging with the latching slot. The structure of the latching slot is simple and easy to produce, which helps reduce manufacturing costs.

[0059] As shown in FIG. **1**, in some embodiments, the quantity of the locking member **20** and the unlocking member **30** is both two. One locking member **20** and one unlocking member **30** are disposed on one side of the supporting member **10**, and the other locking member **20** and the other unlocking member **30** are disposed on the other side of the supporting member **10**. Thus, both sides of the supporting member **10** are connected to the external element **300**, thereby enhancing the stability of the connection between the mounting bracket **100** and the external component **300**.

[0060] As shown in FIGS. **6** and **7**, an embodiment of the present application also provides a server **1000**, including a cabinet **301** (only the portion of the cabinet **301** connected to the mounting bracket **100** is shown in FIGS. **6** and **7**), a fan module **201**, and the mounting bracket **100** according to any of the aforementioned embodiments. The fan module **201** is disposed on the supporting member **10**, and the locking member **20** is connected to the cabinet **301** through a snap-fit manner. The fan module **201** is connected to the cabinet **301** through the mounting bracket **100**, and the assembly and disassembly among the cabinet **301**, the fan module **201**, and the mounting bracket **100** do not require external tools. The operation is simple, which helps improve the efficiency of connecting and disconnecting between the fan module **201** and the cabinet **301**, thereby reducing the assembly and maintenance costs of the server **1000**.

[0061] The above descriptions are some specific embodiments of the present application, but the actual application process cannot be limited only to these embodiments. For those of ordinary skill in the art, other modifications and changes made according to the technical concept of the present application should all belong to the protection scope of the present application.

Claims

1. A mounting bracket comprising: a supporting member configured for supporting a workpiece; a locking member connected to the supporting member, wherein the locking member is configured to

connect to an external component through a snap-fit manner; and an unlocking member slidably connected to the supporting member, wherein the unlocking member is configured to move between a locked position and an unlocked position, such that the locking member deforms, when the unlocking member is in the locked position, the locking member is engaged to the external component, and when the unlocking member is in the unlocked position, the locking member is disengaged from the external component.

2. The mounting bracket of claim 1, further comprising: a handle rotatably connected to the unlocking member, wherein the handle comprises an arcuate surface in contact with the supporting member, the arcuate surface comprises a first contact area and a second contact area, a radius of curvature of the arcuate surface changes from the first contact area to the second contact area, and a radius of curvature of the first contact area is greater than a radius of curvature of the second contact area, the handle has a first positional state where the first contact area is in contact with the supporting member and a second positional state where the second contact area is in contact with the supporting member, the handle is configured to rotate between the first positional state and the second positional state, and when the handle is in the first positional state, the unlocking member is in the unlocked position; when the handle is in the second positional state, the unlocking member is in the locked position.

3. The mounting bracket of claim 2, further comprising: an elastic member **60** connected to the handle and the supporting member, wherein when the handle is in the first positional state, the elastic member has a first deformation amount, when the handle is in the second positional state, the elastic member has a second deformation amount, and the first deformation amount is greater than the second deformation amount.

4. The mounting bracket of claim 2, wherein: the handle comprises a connecting portion, the arcuate surface is formed on the connecting portion; and one of the connecting portion and the supporting member defines a guide groove, another of the connecting portion and the supporting member comprises a guide protrusion, and the guide groove receives the guide protrusion limit an axial movement of the handle.

5. The mounting bracket of claim 1, wherein: the locking member defines a mounting hole, the unlocking member further comprises a protrusion **31**, and the protrusion is configured to be received in the mounting hole; when the unlocking member moves to the unlocked position, the protrusion is configured to move towards an exterior of the mounting hole, such that the locking member deforms; and when the unlocking member moves to the locked position, the protrusion is further configured to move into the mounting hole.

6. The mounting bracket of claim 1, wherein: the unlocking member defines a mounting hole **22**, the locking member comprises a protrusion, the protrusion is configured to be received in the mounting hole; when the unlocking member moves to the unlocked position, the protrusion is configured to move towards an exterior of the mounting hole, such that the locking member deforms; and when the unlocking member moves to the locked position, the protrusion is further configured to move into the mounting hole.

7. The mounting bracket of claim 1, wherein the supporting member comprises a guide post, the unlocking member defines a guide hole, the guide post extends through the guide hole, such that the unlocking member is slidably connected to the supporting member.

8. The mounting bracket of claim 1, wherein the unlocking member comprises a guide post, the supporting member defines a guide hole, the guide post extends through the guide hole, such that the unlocking member is slidably connected to the supporting member.

9. The mounting bracket of claim 1, wherein the supporting member defines a positioning hole, the external component comprises a positioning post, and the positioning post is configured to engage with the positioning hole.

10. The mounting bracket of claim 1, wherein the external component defines a positioning hole, the supporting member comprises a positioning post, and the positioning post is configured to

engage with the positioning hole.

11. The mounting bracket of claim 1, further comprising: a fixing member connected to the supporting member, wherein the fixing member comprises a limiting portion, the limiting portion is configured to connect to a top surface of the workpiece, such that the workpiece presses against the supporting member.

12. The mounting bracket of claim 1, wherein the supporting member defines an installation groove, and the workpiece is at least partially accommodated in the installation groove.

13. The mounting bracket of claim 12, wherein the supporting member comprises a base and a mounting portion, the installation groove is defined on the base, the base is configured to support the workpiece, and the locking member and the unlocking member are configured to install on the mounting portion.

14. The mounting bracket of claim 1, further comprising two locking members and two unlocking members, wherein one of the two locking members and one of the two unlocking members are arranged on one side of the supporting member, and the other of the two locking members and the other of the two unlocking members are arranged on another side of the supporting member.

15. A server comprising: a mounting bracket comprising: a supporting member; a locking member connected to the supporting member, wherein the locking member is connected to an external component through a snap-fit manner; and an unlocking member slidably connected to the supporting member, wherein the unlocking member is configured to move between a locked position and an unlocked position, such that the locking member deforms, when the unlocking member is in the locked position, the locking member is engaged to the external component, and when the unlocking member is in the unlocked position, the locking member is disengaged from the external component; a cabinet connected to the locking member through a snap-fit manner, wherein the cabinet is the external component; and a workpiece arranged on the supporting member.

16. The server of claim 15, wherein: the mounting bracket further comprises a handle rotatably connected to the unlocking member, the handle comprises an arcuate surface in contact with the supporting member, the arcuate surface comprises a first contact area and a second contact area, a radius of curvature of the arcuate surface gradually changes from the first contact area to the second contact area, and a radius of curvature of the first contact area is greater than a radius of curvature of the second contact area, the handle has a first positional state where the first contact area is in contact with the supporting member and a second positional state where the second contact area is in contact with the supporting member, the handle is configured to rotate between the first positional state and the second positional state, and when the handle is in the first positional state, the unlocking member is in the unlocked position; when the handle is in the second positional state, the unlocking member is in the locked position.

17. The server of claim 16, wherein the mounting bracket further comprises an elastic member connected to the handle and the supporting member, when the handle is in the first positional state, the elastic member has a first deformation amount, when the handle is in the second positional state, the elastic member has a second deformation amount, and the first deformation amount is greater than the second deformation amount.

18. The server of claim 16, wherein: the handle comprises a connecting portion, the arcuate surface is formed on the connecting portion; and one of the connecting portion and the supporting member defines a guide groove, another of the connecting portion and the supporting member comprises a guide protrusion, and the guide protrusion is received in the guide groove to limit an axial movement of the handle.

19. The server of claim 15, wherein: the locking member defines a mounting hole, the unlocking member further comprises a protrusion, and the protrusion is configured to be received in the mounting hole; when the unlocking member moves to the unlocked position, the protrusion is configured to move towards an exterior of the mounting hole, such that the locking member

deforms; and when the unlocking member moves to the locked position, the protrusion is further configured to move into the mounting hole.

20. The server of claim 15, wherein: the unlocking member defines a mounting hole, the locking member comprises a protrusion, the protrusion is configured to be received in the mounting hole; when the unlocking member moves to the unlocked position, the protrusion moves towards an exterior of the mounting hole, such that the unlocking member deforms; and when the unlocking member moves to the locked position, the protrusion is further configured to move into the mounting hole.
